SIMATS ENGINEERING

NATURAL LANGUAGE PROCESSING – ASSIGNMENT – 04

Part I

Semantic Analysis Through Structured Reasoning

Details	**Information**
Course	Natural Language Processing
Course Outcome	CO 4 – Analyse and apply semantic representation and interpretation techniques in NLP
Nature of Assignment	Individual
Duration	One Week
Total Marks	50
Submission Date	05-11-26

Student Details

Field	**Details**
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Year / Section	2024
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Task 1 – Meaning Representation and Language Structure (8 Marks)

Scenario: A conversational assistant receives the sentence "Maya gave Ravi a book yesterday." The system must represent who did what, to whom, and when.

1.1 Identify Meaning Components

Component	**Analysis**
Entities	Maya – giver/Agent; Ravi – recipient/Goal; a book – transferred object/Theme.
Event	A giving/transfer event.
Participants	Maya, Ravi, and the book participate in the event.
Semantic roles	Maya = Agent; Ravi = Recipient/Goal; book = Theme.
Temporal information	Yesterday = time of the giving event.

Linguistic form	Subject + Verb + Indirect Object + Direct Object + Temporal Adverbial.
Meaning	There exists a past giving event involving Maya, Ravi, and a book, occurring yesterday.

1.2 Compare Meaning Representations

The same meaning can be represented at different levels. A conceptual/event representation focuses on the event and its participants, while a frame representation organizes the information into named slots.

A simple event structure:

GIVE(e1) ; AGENT(e1, Maya) ; RECIPIENT(e1, Ravi) ; THEME(e1, Book) ; TIME(e1, Yesterday)

Structured frame:

Slot	**Value**
Frame	GIVING
Event ID	e1
Agent	Maya
Recipient	Ravi
Theme	Book
Time	Yesterday

The event structure makes relations explicit and is convenient for reasoning. The frame makes the role slots easy to access in a software system. Both leave some details implicit, such as the exact identity of the book and the precise clock time.

1.3 Meaning Structure and Ambiguity

Word order helps a system identify grammatical roles, but semantic roles are not determined only by word position.

Original:

Maya gave Ravi a book yesterday.

Alternative word order:

Yesterday, Maya gave a book to Ravi.

Both have approximately the same semantic representation:

GIVE(e1)

Agent(e1, Maya)

Recipient(e1, Ravi)

Theme(e1, Book1)

Time(e1, Yesterday)

The words are arranged differently, but the underlying semantic relationships remain equivalent.

1.4 Design Defence

A structured event/frame representation is suitable for a computer because entities, roles and relations are stored as explicit fields rather than hidden in a paraphrase. The representation supports querying, reasoning, information extraction and retrieval. For example, the system can directly answer who received the book or when the event happened. It can also be extended with confidence, entity identifiers and additional relations.

Task 2 – First Order Predicate Calculus (8 Marks)

2.1 Define the Logical Vocabulary

Constants

- Rao – Dr. Rao
- NLP – a particular course

Variables

- x, y – students, instructors or courses depending on context

Symbol	Type / Meaning	Argument Order
Student(x)	x is a student	Unary
Course(c)	c is a course	Unary
Instructor(i)	i is an instructor	Unary
Enrolled(x,c)	student x is enrolled in course c	Student, Course
Teaches(i,c)	instructor i teaches course c	Instructor, Course
Passed(x,c)	student x passed course c	Student, Course
NLP	Constant denoting the NLP course	Constant
DrRao	Constant denoting Dr. Rao	Constant

2.2 Translate Natural Language into Logic

(i) Every student is enrolled in at least one course.

$\forall x\ [Student(x) \rightarrow \exists y\ (Course(y) \wedge Enrolled(x,y))]$

Meaning: For every student, there exists at least one course in which that student is enrolled.

(ii) Some student passed every course taught by Dr. Rao.

$\exists x\ [Student(x) \wedge$

$\quad \forall y\ ((Course(y) \wedge Teaches(Rao,y)) \rightarrow Passed(x,y))]$

Meaning: There is at least one student who passed every course taught by Dr. Rao.

(iii) No instructor teaches every course.

$\forall x\ [Instructor(x) \rightarrow$

$\quad \neg\forall y\ (Course(y) \rightarrow Teaches(x,y))]$

An equivalent form is:

$\forall x\ [Instructor(x) \rightarrow$

$\quad \exists y\ (Course(y) \wedge \neg Teaches(x,y))]$

The second form directly says that every instructor has at least one course that they do not teach.

2.3 Interpret and Check the Formulae

Consider:

$\exists x\ [Student(x) \wedge$

$\quad \forall y\ ((Course(y) \wedge Teaches(Rao,y)) \rightarrow Passed(x,y))]$

Scope

The existential quantifier $\exists x$ has scope over the complete formula inside it.

The universal quantifier $\forall y$ has scope over:

$(Course(y) \wedge Teaches(Rao,y)) \rightarrow Passed(x,y)$

In simple language:

There is one particular student who passed every course taught by Rao.

Quantifier Order

Compare:

$\exists x \forall y$ Passed(x,y)

with:

$\forall y \exists x$ Passed(x,y)

The first means:

There is one student who passed every course.

The second means:

For every course, there may be a different student who passed it.

Counterexample Model

Suppose there are two students:

Alice

Bob

and two courses:

NLP

DBMS

Suppose:

Alice passed NLP

Bob passed DBMS

Then:

$\forall y \exists x$ Passed(x,y)

can be true because every course has some student who passed it.

But:

$\exists x \forall y$ Passed(x,y)

is false because neither Alice nor Bob passed both courses.

2.4 Logical Consistency

Statements:

Every student passed NLP.

There is a student who did not pass NLP.

The formulas are:

$\forall x$ [Student(x) $\rightarrow$ Passed(x,NLP)]

and

$\exists x$ [Student(x) $\wedge \neg$Passed(x,NLP)]

They **cannot both be true**.

The second statement provides a student who did not pass NLP, contradicting the first statement that every student passed NLP.

Task 3 – Linguistically Relevant Concepts and Syntax-Driven Semantic Analysis (9 Marks)

Sentence: "The experienced researcher saw the student with the telescope."

3.1 Identify Linguistically Relevant Concepts

Concept	**Representation / Role**
Researcher	Entity; experienced modifier applies to researcher.
Student	Entity; object of saw.
Telescope	Entity used as an instrument or possible accompaniment.
Saw	Perception event; researcher is Experiencer and student is perceived entity.
With the telescope	PP with ambiguous attachment.
Semantic relations	SEE(e, researcher, student); INSTRUMENT(e,telescope) in the instrument reading.

Important concepts include:

Researcher – subject/experiencer-like participant

Student – object/theme of seeing

Telescope – possible instrument

Experienced – adjective modifying researcher

Saw – perception event

These concepts are necessary because the same words can produce different meanings depending on their semantic relationships.

3.2 Syntax-to-Semantics Mapping

Attachment A – VP attachment: "The experienced researcher saw [the student] [with the telescope]." The PP modifies the seeing event. Semantic reading: the researcher used the telescope while seeing the student.

SEE(e1, Researcher, Student) ∧ INSTRUMENT(e1, Telescope)

Attachment B – NP attachment: "The experienced researcher saw [the student with the telescope]." The PP modifies the noun phrase student. Semantic reading: the student is associated with/has the telescope.

SEE(e1, Researcher, Student) ∧ WITH(Student, Telescope)

Syntax-driven procedure: build the parse tree → identify the head of each phrase → compose the semantic representation of the head with its complements/modifiers → produce candidate sentence meanings → compare candidates using context.

3.3 Semantic Composition

The interpretation can be built in stages.

Step 1 – Noun meanings

researcher → RESEARCHER(x)

student → STUDENT(y)

telescope → TELESCOPE(z)

Step 2 – Adjective

experienced researcher

becomes:

RESEARCHER(x) ∧ EXPERIENCED(x)

Step 3 – Verb

saw

introduces a seeing event:

SEE(e,x,y)

Step 4 – Object

the student

fills the Theme/Object role:

Theme(e,y)

Step 5 – PP

If attached to VP:

Instrument(e,z)

If attached to NP:

HAS(y,z)

Therefore, the interpretations diverge at the **attachment of the PP**.

3.4 Failure Analysis

Error	Consequence	Validation Check
Wrong PP attachment	System may claim the student owns/has the telescope when it was actually used as an instrument.	Check event-role compatibility and use context such as "looked through the telescope."
Assigning roles from word order alone	A PP or passive construction may be assigned the wrong semantic role.	Use dependency/constituency structure plus a role-compatibility check against the verb's frame.

Task 4 – Lexemes, Senses and Word Sense Disambiguation (9 Marks)

4.1 Lexemes and Senses

A lexeme is an abstract vocabulary item associated with a family of word forms and senses. A word form is the actual surface form appearing in text. A sense is a particular meaning of a lexeme in context.

Context	Word form	Lexeme / grammatical use	Relevant sense
A: Nila deposited cash	bank	BANK noun	Financial institution

at the bank.			
B: They sat on the bank of the river.	bank	BANK noun	River edge / riverside
C: The pilot banked the aircraft to the left.	banked	BANK verb	Tilt/turn an aircraft laterally

The noun uses in A and B share the word form “bank” but have different senses. C is a past-tense verb form of the verbal lexeme BANK.

4.2 Design a Context-Based WSD Method

Input

A sentence containing the ambiguous word bank.

Output

The most probable sense of bank.

Candidate Senses

S1 = Financial institution

S2 = River edge

S3 = Aircraft movement/tilting

Context Features

Use:

Words occurring near bank

Part of speech

Verb-object relationships

Semantic category of nearby nouns

Prepositions

Domain indicators

Decision Procedure

Step 1: Identify the target word.

Step 2: Determine its part of speech.

Step 3: Generate candidate senses.

Step 4: Extract nearby context words.

Step 5: Compare context with the semantic features of each sense.

Step 6: Select the sense with the highest compatibility score.

Example:

cash + deposited + bank

strongly supports the financial sense.

4.3 Test the Method

Sentence	**Evidence**	**Selected Sense**
Nila deposited cash at the bank.	deposited, cash, at; financial vocabulary	FINANCIAL
They sat on the bank of the river.	sat, river, on the bank	RIVER-EDGE
The pilot banked the aircraft to the left.	pilot, aircraft, banked, left	AIRCRAFT-TURN
The company asked for a bank loan.	company, loan, financial context	FINANCIAL

Possible failure: “The bank was crowded” is ambiguous without context because both a financial institution and a river bank can be crowded. A larger context window or a knowledge-based clue is needed.

4.4 Evaluation

Create a labelled test set containing sentences for each sense, for example 30 sentences with 10 examples per sense. Compare predicted labels with gold labels.

Accuracy = correctly classified examples / total examples.

If 27 of 30 examples are correct, accuracy = 27/30 = 90%. A balanced test set can hide real-world weaknesses because actual corpora may have highly unequal sense frequencies, domain shifts, rare

senses and new contexts. Therefore, per-sense recall, confusion matrices and domain-specific evaluation should also be reported.

Task 5 – Information Retrieval and Semantic Matching (8 Marks)

5.1 Query and Document Representation

Query: "methods for finding the meaning of ambiguous words."

Element	Semantic interpretation
Central concept 1	Word meaning / lexical semantics
Central concept 2	Ambiguity / multiple senses
Central concept 3	Methods for disambiguation
Likely intent	Find techniques or algorithms for determining the intended sense of an ambiguous word.
Query representation	Concept vector / embedding + extracted concepts such as WSD, lexical ambiguity, word senses.
Document representation	Title, abstract, content concepts, embeddings, keywords and domain metadata.

5.2 Lexical versus Semantic Retrieval

Method	Strength	Weakness
Keyword matching	Simple, transparent and efficient when query and document share terms.	May miss documents using synonyms or different terminology.
Semantic retrieval	Matches concepts and related meanings even when exact words differ.	May retrieve related but non-answering documents if semantic similarity is too broad.

Example missed by keyword retrieval: a document titled "Determining the Intended Sense of Polysemous Terms Using Context" may discuss WSD without using the exact phrase "meaning of ambiguous words." Semantic retrieval can connect polysemy, context and sense selection.

5.3 Ranking Design

Use a weighted ranking score:

$$\text{Score}(d,q) = 0.35 \times \text{TermRelevance} + 0.35 \times \text{ConceptSimilarity} + 0.20 \times \text{SenseMatch} + 0.10 \times \text{FieldWeight}.$$

Signal	Effect
Term relevance	Rewards important query terms and avoids overvaluing common words.
Concept similarity	Rewards semantic closeness between query and document.
Sense match	Rewards documents explicitly addressing ambiguity and word senses.
Field weighting	Gives additional importance to title/abstract matches.

5.4 Evaluation and Error Analysis

Precision = relevant retrieved documents / all retrieved documents. Recall = relevant retrieved documents / all relevant documents.

Document	Retrieved?	Relevant?	Classification
D1 – WSD methods	Yes	Yes	TP
D2 – lexical semantics	Yes	Yes	TP
D3 – general NLP introduction	Yes	No	FP
D4 – information retrieval	No	Yes	FN
D5 – machine translation	No	No	TN

If 2 relevant documents are retrieved out of 3 retrieved documents, Precision = 2/3 = 66.7%. If there are 3 relevant documents in the collection and 2 are retrieved, Recall = 2/3 = 66.7%.

A semantic retrieval error occurs when a document discusses “word embeddings” and is highly similar to the query but does not actually explain methods for disambiguating word senses. A sense-aware relevance check can reduce this error.

Task 6 – Integrated Semantic Analysis Challenge (8 Marks)

Message: “I saw her duck near the bank, and it was closed.”

This sentence contains several ambiguities.

Stage 1 – Interpret the Problem

Ambiguity 1: duck

It can be:

A noun: an animal

A verb: lower one's head/body

Ambiguity 2: bank

It can mean:

Financial institution

River bank

Ambiguity 3: it

The pronoun may refer to:

The bank

Another previously mentioned entity

Ambiguity 4: closed

Most naturally describes a business/place such as a bank, but context is needed.

Stage 2 – Build Candidate Interpretations

Interpretation 1 – Duck as an Animal

Meaning:

I saw her duck near the river bank, and the river bank was closed.

Semantic representation:

SEE(e1)

Agent(e1, Speaker)

Theme(e1, Duck1)

Location(e1, NearBank1)

DUCK(Duck1)

RIVER_BANK(Bank1)

CLOSED(Bank1)

Possible interpretation:

Speaker saw a duck near a river bank.

The river bank was closed.

However, the pronoun resolution depends heavily on context.

Interpretation 2 – Duck as a Verb

Meaning:

I saw her lower herself near a bank, and the bank was closed.

Representation:

SEE(e1)

Agent(e1, Speaker)

Experiencer(e1, Her)

DUCK(e2)

Agent(e2, Her)

Location(e2, NearBank1)

BANK(Bank1)

CLOSED(Bank1)

Here duck functions as a verb, describing an action performed by her.

If bank refers to a financial institution, the interpretation could be:

I saw her duck near the bank, and the bank was closed.

This is quite plausible in a real-world context.

Stage 3 – Syntax-Driven Analysis and WSD

The system should first examine possible syntactic structures.

Candidate A

her duck

is treated as a noun phrase:

her + duck(noun)

Meaning:

the duck belonging to/associated with her.

Candidate B

her duck

is interpreted with duck as a verb:

her + duck(verb)

Meaning:

she lowers herself.

The system then uses context.

For bank

If nearby context contains:

money

account

loan

cash

ATM

select:

BANK = financial institution

If it contains:

river

water

shore

fishing

select:

BANK = river edge

Most Useful Additional Context

The best additional information would be:

Previous sentence

Following sentence

Location

Whether the conversation concerns banking

Whether a duck/animal has already been mentioned

Whether a bank is physically nearby

Stage 4 – Information Retrieval Trace

Selected interpretation: bank = financial institution; duck = verb or bird depending on the syntactic parse; "closed" is a strong clue that the bank institution is the relevant entity.

1. Extract concepts: bank closure, financial institution, opening hours, branch access.
2. Normalize terms: closed → closure/closed branch; bank → financial institution.
3. Construct semantic query: {financial institution, bank branch, closed, opening hours, access} plus context embeddings.
4. Retrieve candidate articles using lexical and semantic similarity.
5. Apply sense filter: reject documents about river banks unless other evidence supports that sense.
6. Rank remaining documents using term relevance, concept similarity and sense match.

Example ranking:

Article	**Bank Sense**	**Topic Match**	**Rank Decision**
Branch closed – opening hours	Financial	High	1
Why river banks flood	River	Low	Reject
Bank holiday schedule	Financial	High	2

Stage 5 – Challenge and Refine

Misleading Context

Suppose the previous message says:

"We went for a walk beside the river."

The system may incorrectly choose:

bank = river bank

because river is strongly associated with that sense.

However, the phrase:

“it was closed”

may indicate a bank branch, especially if the conversation is about banking services.

Refinement

Add cross-sentence context and semantic-role compatibility.

The system should combine:

Local context

+

Previous sentences

+

Syntactic structure

+

World knowledge

+

Sense compatibility

instead of relying on a single nearby word.

Conclusion

The assignment demonstrates that semantic analysis is more than replacing words with dictionary definitions. A robust NLP system must connect linguistic structure with explicit meaning representations, logical reasoning, word-sense decisions and retrieval objectives. Event structures and frames provide machine-readable semantic roles; FOPC supports quantifier-based reasoning; syntax-driven composition handles attachment ambiguity; WSD separates lexemes from context-specific senses; and semantic retrieval connects user intent to documents even when wording differs.

For the integrated helpdesk scenario, the strongest design is a pipeline that combines syntactic parsing, compositional semantics, WSD, coreference resolution and sense-aware retrieval. The system should preserve alternative interpretations when evidence is insufficient rather than silently selecting an unreliable meaning.

SIMATS ENGINEERING

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Part II

SEMANTIC SYSTEMS ENGINEERING: ANALYSE, REPRESENT, TEST, DISAMBIGUATE

& RETRIEVE

Details	**Information**
Course	Natural Language Processing
Course Outcome	CO 4 – Analyse semantic structure and design meaning-based NLP systems
Nature of Assignment	Individual
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Name	G. Sirisha
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TASK 1 – REPRESENTING MEANING AND LANGUAGE STRUCTURE (8 Marks)

Sentence: “Leena sent the report to Arun after the meeting.”

1.1 Semantic Structure Analysis | 2 Marks

Semantic inventory:

Component	**Representation**
Event	Sending
Agent	Leena

Theme	Report
Recipient	Arun
Temporal expression	After the meeting
Event time	After meeting
Implicit information	The report was transferred/sent to Arun.

A semantic representation is:

Event: e1

Type: SEND

Agent: Leena

Theme: Report1

Recipient: Arun

Time: After(Meeting1)

Surface form: Leena → sent → the report → to Arun → after the meeting.

Meaning representation: SEND(e1), AGENT(e1, Leena), THEME(e1, Report1), RECIPIENT(e1, Arun), AFTER(e1, Meeting1).

1.2 Build Meaning Representations | 3 Marks

A. Semantic-role/event frame

Role	**Value**
Agent	Leena
Theme	Report
Recipient	Arun
Time relation	After meeting

SEND(e1)

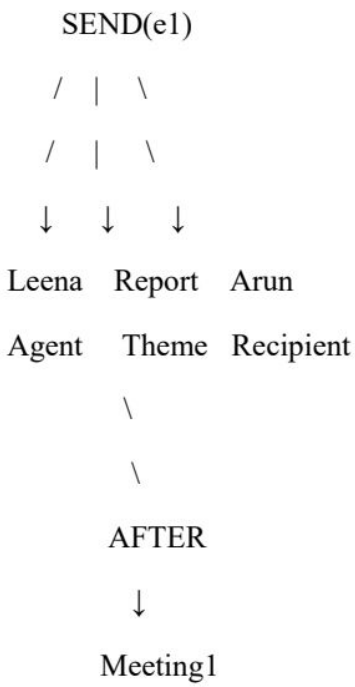
AGENT(e1, Leena)

THEME(e1, Report1)

RECIPIENT(e1, Arun)

AFTER(e1, Meeting1)

B. Graph representation



Graph triples:

(e1, type, SEND)

(e1, agent, Leena)

(e1, theme, Report1)

(e1, recipient, Arun)

(e1, temporal_relation, AFTER)

(e1, reference_event, Meeting1)

1.3 Meaning-Preserving Variation | 2 Marks

Original:

Leena sent the report to Arun after the meeting.

Alternative:

After the meeting, Leena sent Arun the report.

The surface order changes, but the semantic relations remain:

AGENT(e1, Leena)

THEME(e1, Report1)

RECIPIENT(e1, Arun)

AFTER(e1, Meeting1)

Meaning-changing variation

Leena sent Arun's report to Ravi after the meeting.

Now the theme is different:

THEME(e1, Report2)

RECIPIENT(e1, Ravi)

Therefore, changing syntactic structure does not necessarily change meaning, but changing semantic role assignments or entity relationships does.

1.4 Representation Check | 1 Mark

A validation rule can check whether every SEND event has a required Agent, Theme and Recipient.

IF Event = SEND

THEN Agent ≠ NULL

Theme ≠ NULL

Recipient ≠ NULL

For example:

SEND(e1)

AGENT(e1, Leena)

THEME(e1, Report1)

RECIPIENT(e1, Arun)

passes the validation.

If:

RECIPIENT(e1, NULL)

the system reports:

Error: Missing recipient for SEND event.

TASK 2 – FIRST ORDER PREDICATE CALCULUS: FORMALISE AND TEST (9 Marks)

2.1 Predicate Vocabulary | 2 Marks

Constants

NLP = NLP course

CS = Computer Science department

Variables

x, y, z = learners/courses

Symbol	Meaning	Argument Order
Learner(x)	x is a learner	entity
Course(x)	x is a course	entity
Enrolled(x,y)	learner x is enrolled in course y	learner, course
Completed(x,y)	learner x completed course y	learner, course
Prerequisite(x,y)	x is a prerequisite of y	prerequisite course, target course
OfferedBy(x,y)	course x is offered by department y	course, department

Constants: NLP = NLP course; CS = Computer Science department. Variables x, y, z represent learners, courses or departments according to context.

2.2 Formalisation | 3 Marks

(i) Every learner enrolled in NLP has completed at least one prerequisite.

$$\forall x \, [$$

$$\text{Learner}(x) \land \text{Enrolled}(x,\text{NLP})$$

$$\rightarrow$$

$$\exists y \, [$$

$$\text{Course}(y) \land$$

Prerequisite(y,NLP) ∧

Completed(x,y)

]

]

Meaning:

Every learner enrolled in NLP has completed at least one course that is a prerequisite of NLP.

(ii) Some learner completed every course offered by the department.

Assume:

OfferedBy(y, CS)

means course y is offered by the department.

Then:

∃x [

Learner(x) ∧

∀y [

(Course(y) ∧ OfferedBy(y,CS))

→

Completed(x,y)

]

]

Meaning:

There exists a learner who completed every course offered by the department.

(iii) No learner completed a course that they were not enrolled in.

∀x∀y [

(Learner(x) ∧ Course(y) ∧ Completed(x,y))

→

Enrolled(x,y)

]

Equivalent:

∀x∀y [

Learner(x) ∧ Course(y)

→

(Completed(x,y) → Enrolled(x,y))

]

2.3 Quantifier-Scope Challenge | 2 Marks

Consider:

∃x ∀y Completed(x,y)

This means:

There is one learner who completed every course.

Compare:

∀y ∃x Completed(x,y)

This means:

For every course, there is some learner who completed it.

The two formulas are **not logically equivalent**.

Finite Model

Learners:

Alice

Bob

Courses:

NLP

DBMS

Facts:

Completed(Alice,NLP)

Completed(Bob,DBMS)

Then:

∀y ∃x Completed(x,y)

is true because:

- NLP → Alice completed it.
- DBMS → Bob completed it.

But:

∃x ∀y Completed(x,y)

is false because neither Alice nor Bob completed both courses.

Therefore:

Quantifier order changes the meaning.

2.4 Consistency Test | 2 Marks

Knowledge Base

Facts:

Learner(Asha)

Course(NLP)

Course(DBMS)

Enrolled(Asha,NLP)

Completed(Asha,NLP)

Rule:

∀x∀y [

Completed(x,y) → Enrolled(x,y)

]

Query

Enrolled(Asha,NLP)

Inference

The fact is directly present:

Enrolled(Asha,NLP)

Therefore:

Query follows from the knowledge base.

Counterexample Query

Consider:

Enrolled(Asha,DBMS)

There is no fact or rule establishing this.

Therefore:

The query cannot be derived from the given knowledge base.

This demonstrates the difference between **being unsupported** and being logically contradicted.

TASK 3 – LINGUISTIC CONCEPTS AND SYNTAX-DRIVEN SEMANTIC ANALYSIS (9 Marks)

Sentence: "The analyst noticed the student with the camera near the library."

3.1 Linguistically Relevant Concepts | 2 Marks

Concept	Example / Role
Entity 1	Analyst
Entity 2	Student
Instrument	Camera
Location	Library
Predicate	Noticed
PP 1	with the camera
PP 2	near the library

Possible attachment	Student / noticing event

These concepts matter because prepositional phrases can attach to different constituents and therefore create different semantic relations.

3.2 Parse-to-Meaning Procedure | 3 Marks

Interpretation 1 – Camera as Instrument of Noticing

Meaning:

The analyst used a camera while noticing the student near the library.

Representation:

NOTICE(e1)

Observer(e1, Analyst)

Theme(e1, Student)

Instrument(e1, Camera)

Location(e1, Library)

Possible structure:

S

├── NP

│ └── The analyst

└── VP

├── V

│ └── noticed

├── NP

│ └── the student

└── PP

├── with the camera

└── near the library

Composition

The analyst

↓

ANALYST(x)

the student

↓

STUDENT(y)

noticed

↓

NOTICE(e,x,y)

with the camera

↓

INSTRUMENT(e,Camera)

near the library

↓

LOCATION(e,Library)

Final meaning:

NOTICE(e1)

Observer(e1, Analyst)

Theme(e1, Student)

Instrument(e1, Camera)

Location(e1, Library)

Interpretation 2 – Camera Modifies the Student

Meaning:

The analyst noticed the student who had/was associated with the camera near the library.

Representation:

NOTICE(e1)

Observer(e1, Analyst)

Theme(e1, Student)

HAS(Student, Camera)

Location(Student, Library)

Here:

with the camera

is attached to the noun phrase:

the student

rather than directly to the noticing event.

Exact Point of Divergence

The interpretations diverge at:

with the camera

Interpretation 1

with camera → NOTICE event

Interpretation 2

with camera → STUDENT

Therefore, the syntactic attachment determines the semantic role.

3.3 Composition and Type/Role Checks | 2 Marks

Step 1

analyst → ANALYST(x)

student → STUDENT(y)

camera → CAMERA(z)

library → LIBRARY(l)

Step 2

The verb introduces an event:

NOTICE(e,x,y)

Step 3

The PP:

with the camera

can produce:

INSTRUMENT(e,z)

or:

HAS(y,z)

Step 4

The location phrase contributes:

AT(e,l)

or:

NEAR(y,l)

depending on attachment.

Semantic Check 1 – Selectional Preference

Check whether:

NOTICE + Instrument(Camera)

is semantically plausible.

If the system knows that cameras can be instruments for visual observation, this interpretation receives support.

Semantic Check 2 – Type Compatibility

Check whether the selected semantic relation accepts the argument type.

For example:

INSTRUMENT(event, Camera)

is valid.

But:

INSTRUMENT(event, Library)

would be suspicious.

3.4 Error Challenge | 2 Marks

Case

The analyst noticed the student with the camera near the library.

Suppose the parser attaches:

with the camera

to student.

The system may conclude:

The student possessed the camera.

However, the actual context may be:

The analyst used the camera to observe the student.

Predicted Wrong Output

HAS(Student, Camera)

Refinement

Use:

1. Context
2. Selectional preferences
3. Verb-frame knowledge
4. Previous and following sentences

For example, if the previous sentence says:

The analyst was conducting a photography survey.

then:

Camera → Instrument of noticing

becomes more probable.

TASK 4 – LEXEMES, SENSES AND WORD SENSE DISAMBIGUATION (8 Marks)

Target lexeme: “light”

4.1 Lexeme and Sense Inventory | 2 Marks

Sense	Meaning	POS	Example
S1	Lamp / lighting device	Noun	Turn on the light.
S2	Not heavy	Adjective	This bag is light.
S3	Low in richness/amount or calories	Adjective	The light meal was served early.
S4	Illumination / radiance	Noun	The light from the window was bright.

A lexeme is an abstract vocabulary unit, a word form is its surface realization, and a sense is one specific meaning of the lexeme.

4.2 WSD Design | 2 Marks

Input

A sentence containing:

light

Candidate Senses

S1 = lamp

S2 = not heavy

S3 = low in quantity/calories

S4 = illumination

Contextual Features

Use:

- Part of speech
- Nearby words
- Verb relationships
- Noun modifiers
- Semantic category of surrounding words

Decision Procedure

1. Locate "light".

2. Determine its POS.

3. Generate candidate senses.

4. Examine nearby context.

5. Compare context with sense indicators.

6. Select the sense with the highest compatibility.

A simple scoring rule is:

Score(sense) =

ContextMatch

+

POSMatch

+

SemanticCompatibility

The sense with the highest score is selected.

4.3 Apply and Challenge | 2 Marks

Sentence	Evidence	Selected Sense
Turn on the light.	turn on; device-like object	S1 – lamp

This bag is light.	bag; adjective of weight	S2 – not heavy
The light meal was served early.	meal; food description	S3 – light food
The light from the window was bright.	window; bright; illumination	S4 – illumination

Failure case: “The light was good.” The context may not distinguish a lamp from illumination. The system should retain alternatives or use wider context.

4.4 Evaluation | 2 Marks

No.	Sentence	Gold Sense
1	Turn on the light.	S1
2	The light was bright.	S4
3	This bag is light.	S2
4	The light meal was tasty.	S3
5	Switch off the light.	S1
6	This box is light.	S2
7	We had a light lunch.	S3
8	Sunlight entered the room.	S4

If 7 of 8 are correctly classified: Accuracy = 7/8 = 87.5%. Class imbalance can make accuracy misleading; per-sense precision, recall, F1-score and a confusion matrix are also useful.

TASK 5 – SEMANTIC INFORMATION RETRIEVAL (8 Marks)

Query: “How can a system identify the intended meaning of a word in context?”

5.1 Query and Document Semantics | 2 Marks

Intent: Information seeking

Task: Determine word meaning

Context: Surrounding linguistic context

Topic: Word Sense Disambiguation

Key concepts: word, meaning, context, sense, disambiguation.

ID	Candidate document	Type
D1	Context-Based Word Sense Disambiguation Methods	Relevant
D2	Supervised Approaches to Lexical Semantics	Relevant
D3	Dictionary-Based Word Sense Disambiguation	Relevant
D4	Word Frequency and Corpus Statistics	Related / distractor
D5	Meaning of Words in Historical Dictionaries	Related but not directly answering
D6	How to Improve Search Engine Ranking	Distractor

5.2 Ranking Strategy | 2 Marks

$$\text{Score}(D) = 0.4 \times \text{LexicalScore}(D) + 0.6 \times \text{SemanticScore}(D)$$

Tie handling: prefer higher concept coverage; if tied, prefer higher title relevance; if still tied, prefer higher source quality.

5.3 Retrieval Trace | 2 Marks

Document	Lexical	Semantic	Final Score
D1	0.80	0.90	0.86
D2	0.55	0.85	0.73
D3	0.60	0.80	0.72
D4	0.75	0.40	0.54
D5	0.45	0.60	0.51
D6	0.70	0.20	0.50

Example: Score(D1) = 0.4(0.80) + 0.6(0.90) = 0.86. Final ranking: D1 > D2 > D3 > D4 > D5 > D6.

D2 may contain fewer exact query words but rank highly because it is semantically related to lexical semantics and word-sense identification.

5.4 Evaluate | 2 Marks

Suppose we retrieve the top 4:

D1, D2, D3, D4

Assume:

Relevant = D1, D2, D3

Then:

Relevant Retrieved = 3

Retrieved = 4

Total Relevant Documents = 3

Precision

Precision = 3 / 4

= 0.75

= 75%

Recall

Recall = 3 / 3

= 1

= 100%

False Positive

D4

is retrieved but is not sufficiently relevant.

False Negative

Suppose D7 is a highly relevant article about:

Contextual word meaning prediction

but it was not retrieved.

This is a false negative.

Improvement

Increase the weight of semantic similarity and add **concept coverage** to the ranking model.

TASK 6 – INTEGRATED SEMANTIC ANALYSIS DESIGN CHALLENGE (8 Marks)

Sentence: “I watched the crane by the bank, then it moved.”

6.1 Model the Ambiguity | 1 Mark

There are two major lexical ambiguities.

Ambiguity 1 – Crane

crane may mean:

1. A **bird**
2. A **machine used for lifting**

Ambiguity 2 – Bank

bank may mean:

1. Financial institution
2. River edge

Pronoun Ambiguity

The word:

it

could refer to:

- Crane
- Bank

The most natural interpretation is usually **crane**, because a crane is an entity capable of moving.

Required Context

The system needs:

- Surrounding sentences
- Location information
- Nearby words

- Domain
- Verb compatibility
- World knowledge

6.2 Candidate Meaning Structures | 2 Marks

Interpretation 1 – Crane as a Bird, Bank as River Edge

Meaning:

The speaker watched a crane bird near the bank of a river, and the crane moved.

Representation:

WATCH(e1)

Agent(e1, Speaker)

Theme(e1, Crane1)

CRANE_BIRD(Crane1)

RIVER_BANK(Bank1)

NEAR(Crane1, Bank1)

MOVE(e2)

Agent(e2, Crane1)

The pronoun:

it → Crane1

because birds can move.

Interpretation 2 – Crane as Lifting Machine, Bank as Financial Institution

Meaning:

The speaker watched a construction crane near a bank building, and the crane moved.

Representation:

WATCH(e1)

Agent(e1, Speaker)

Theme(e1, Crane2)

CRANE_MACHINE(Crane2)

BANK_BUILDING(Bank2)

NEAR(Crane2, Bank2)

MOVE(e2)

Agent(e2, Crane2)

Here it again most naturally refers to the crane.

6.3 Syntax and WSD Trace | 2 Marks

Step 1 – Identify noun phrase

the crane

Candidate senses:

CRANE_BIRD

CRANE_MACHINE

Step 2 – Identify PP

by the bank

The PP gives a location relation:

NEAR(Crane, Bank)

Step 3 – Determine bank sense

Possible:

BANK_FINANCIAL

BANK_RIVER

Step 4 – Use Context

If the surrounding context contains:

construction

building

workers

lifting

then:

crane → machine

bank → financial institution/building

If it contains:

water

river

birds

wildlife

then:

crane → bird

bank → river edge

Step 5 – Resolve it

Use semantic compatibility:

MOVE(crane)

is natural for both a bird and a machine.

Therefore, pronoun resolution may still require additional context.

Insufficient Evidence

If neither interpretation is sufficiently supported, the system should **retain both interpretations** or ask:

“Do you mean a crane bird or a construction crane?”

This is better than silently selecting an unsupported sense.

6.4 Semantic Retrieval | 2 Marks

Assume the selected interpretation is: crane = construction machine; bank = financial institution/building.

Topic: Construction crane near a bank

Entity: Crane machine

Location: Bank/building

Event: Movement

Document	Description	Lexical	Semantic	Final
D1	Construction Crane Operation and Safety	0.80	0.95	0.89
D2	Crane Maintenance and Movement	0.75	0.90	0.84
D3	Bank Branch Opening Hours	0.70	0.35	0.49
D4	Crane Birds Near River Banks	0.90	0.25	0.51
D5	Financial Bank Services	0.65	0.20	0.38

Score formula: 0.4(Lexical) + 0.6(Semantic). For D1: 0.4(0.80)+0.6(0.95)=0.89. Final ranking: D1 > D2 > D4 > D3 > D5. D4 has strong lexical overlap but a wrong sense, so its semantic score is low.

6.5 Adversarial Refinement | 1 Mark

Misleading Context

Suppose the previous sentence says:

“We visited the wetlands yesterday.”

The system may strongly prefer:

crane = bird

bank = river edge

This could be correct, but another sentence might reveal:

“A construction company was repairing the bank building.”

Now the intended interpretation changes.

Predicted Result

A system using only the previous sentence may select the wrong senses.

Refinement

Use **multi-sentence contextual reasoning** and retain competing hypotheses until enough evidence is available.

The system can use:

Local context

+

Document context

+

Syntactic attachment

+

Selectional preferences

+

World knowledge

+

Pronoun compatibility

This reduces premature WSD decisions.

FINAL INTEGRATED ENGINEERING WORKFLOW

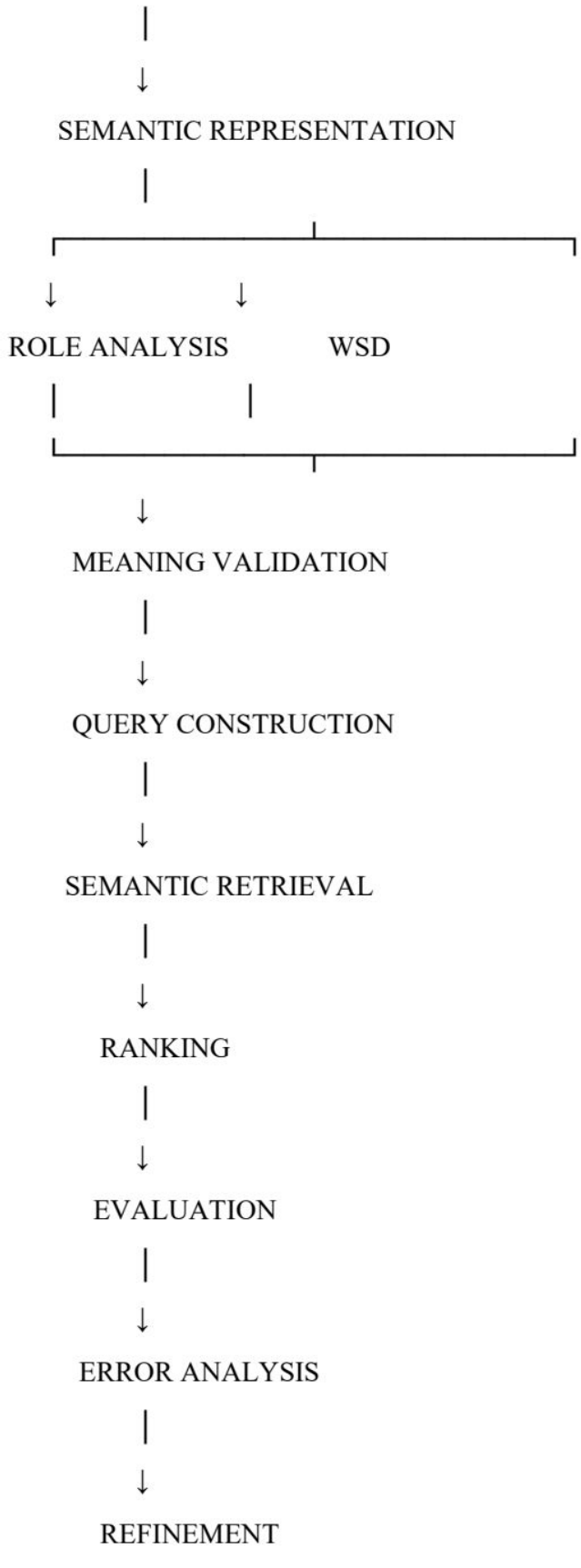
INPUT SENTENCE

↓

LINGUISTIC ANALYSIS

↓

SYNTACTIC PARSING


SEMANTIC REPRESENTATION
ROLE ANALYSIS
WSD
MEANING VALIDATION
QUERY CONSTRUCTION
SEMANTIC RETRIEVAL
RANKING
EVALUATION
ERROR ANALYSIS
REFINEMENT

Conclusion

This assignment demonstrates that a semantic NLP system should not stop after producing an interpretation. The system must represent meaning explicitly, trace how syntax produces that meaning, identify and resolve lexical ambiguity, evaluate retrieval quality, and test the design against misleading inputs. Predicate logic supports formal reasoning, semantic-role and graph representations capture event structure, WSD selects appropriate lexical senses, and semantic retrieval connects the resulting meaning to relevant documents. The final system should preserve alternative interpretations when evidence is insufficient rather than making an unsupported decision.